FE5108 / PRACTITIONER SEGMENT FINAL PROJECT

Factor Investing and Portfolio Analytics

Portfolio Theory and Investments | Fall 2026

National University of Singapore • November 2026

Research overview

We examine factor investing, security-level portfolio rules, industry optimization, daily leveraged products and portfolio attribution. The evidence shows that a shared factor label does not imply identical returns, and that historical optimization gains need validation outside the estimation sample.

Our industry allocation analysis favors long-only minimum variance over unrestricted optimization out of sample. Daily 2× leverage substantially raises risk and does not translate into twice the multi-day stock return. Factor replication explains most MSCI Value variation, while the multifactor index generates only a small active return relative to its tracking error.

Team contributions

Team member	Assigned contribution
Leng Bojun	Part 1 Factor investing
Zhang Boyi	Part 2 Security-level portfolio construction
Ma Yuyang	Part 3 Industry portfolio optimization
Wang Mingyuan	Parts 4 and 5 Product innovation and portfolio analytics

Part 1 Factor investing

1A Historical evidence for factor premia

Factor returns reflect both an investment idea and the rules used to implement it. We first examine the arithmetic mean of the Fama–French market, size and value spreads in two nonoverlapping periods. Ordinary mean t statistics use the monthly sample standard deviation divided by the square root of the observation count; no autocorrelation adjustment is applied. For 1927–1992, the market, SMB and HML t statistics are 3.22, 2.13 and 3.33. For January 1993–July 2023, they are 3.09, 0.60 and 0.83 (Table A1). Only the later market mean remains significant at the conventional two-sided 5% level.

The later results give us less confidence that size and value earn positive average returns. They do not establish that publication eliminated either premium: separate mean-difference tests do not reject equal means for SMB or HML at 5%. For subsequent comparisons we also construct value-weighted highest-minus-lowest momentum decile returns and lowest-minus-highest beta decile returns. These spreads are distinct from total returns on fully invested index portfolios [1].

1B Implementing momentum in an MSCI index

Under the assigned 2013 MSCI method, the universe is USA Standard large/mid-cap constituents. Six-and twelve-month local price gains exclude the latest month, deduct local risk-free rates and are scaled by three-year weekly-return volatility. Signals are standardized, equally combined and standardized again. Final Z scores are capped at ±3 and transformed to $1 + Z$ for $Z > 0$ or $1/(1 - Z)$ otherwise [2].

Selection ranks uncapped Z scores, with parent weight breaking ties. For a reviewed target of N stocks, the top 0.5N ranks enter first; incumbents ranked within 1.5N are retained until N places are filled, with vacancies filled by rank. Weights normalize parent capitalization weights times momentum scores, subject to a 5% USA issuer cap. Capitalization supports trading capacity and limits oversized small-stock positions, but can dilute momentum exposure.

Reviews occur in May and November. Monthly checks trigger extra reviews when the relative increase in trailing three-month daily market volatility exceeds a historical 95th-percentile threshold, using MSCI World as the developed-market reference. Triggered reviews use six-month momentum to respond faster to volatility shocks. These historical rules need not describe every index observation since 1995.

1C Evaluating a diversified factor allocation

We convert MSCI Gross USD month-end levels to monthly total returns and reset Value, Minimum Volatility, Equal Weight and Momentum to 25% each month. The common comparison uses January 1995–July 2023 and 343 observations, with December 1994 retained as the initial index level. Table A2 reports arithmetic annualized means, sample volatility and the assignment Sharpe definition.

Momentum has the highest annual mean, 13.38%, and Value the lowest, 9.87%. Momentum and Minimum Volatility have nearly tied Sharpes of 0.686 and 0.683. The multifactor allocation returns 11.37% with 14.40% volatility and a 0.639 Sharpe, compared with 11.22%, 15.44% and 0.586 for USA. It improves on the market and some components without dominating the best individual Sharpe. Its volatility is below the 15.21% weighted average of component volatilities, the analytical reference under perfect positive correlation

Sources: instructor data and cited methods. Linked results: project repository

(Figure A1). The reduction is 0.81 percentage points, or 5.3% relative to that reference. This isolates a diversification benefit; it does not identify an investable alternative.

1D Sensitivity to portfolio construction

For each MSCI factor, we subtract USA returns before correlating the result with HML, SMB, decile momentum and low-beta spreads. Corresponding correlations are 0.728 for Value– HML, 0.322 for Equal Weight–SMB, 0.643 for Momentum–momentum and 0.815 for Minimum Volatility–low beta (Table A3). The correlations still reflect the intended factors, but equal weighting does not closely match a small-minus-big strategy. Differences in universe, size balance, weighting and the subtracted benchmark explain why similar investment themes need not produce identical spreads.

1E Momentum across market volatility regimes

From December 1995–July 2023, trailing twelve-month Mkt−RF volatility has a negative association with decile momentum returns. The contemporaneous regression slope is −0.267, HAC(12) p = 0.022 and R^2 = 3.2% (Figure A2; Table A4). A ten-percentage-point volatility increase corresponds to 2.67 percentage points less monthly momentum return. Low-, middle-and high-volatility terciles have annualized mean momentum spreads of 12.47%, 22.34% and −12.33%; the relationship is not uniformly decreasing. A lagged-volatility specification explains only 2.2% of variation and is sensitive to extreme months. The relationship is weak in this sample. Momentum performs poorly on average in high-volatility months, but the results do not show a reliable forecasting rule.

Part 2 Security level portfolio construction

2A Validating the stock panel

Construction choices can be examined directly in the anonymized security panel. Across 245 formation months, November 1998–March 2019, all 283,380 recorded stock-month returns are observed. Stock counts fall from 1,109 to 911 in December 2018, a loss of 198 stocks (17.9%), and from 907 to 555 in February 2019, a loss of 352 stocks (38.8%). Both declines are large in absolute and percentage terms, making these months particularly questionable for consistency of universe coverage (Figure A3). Counts alone cannot establish the cause. If returns were missing, we would investigate prices and corporate events rather than exclude holdings using unavailable future outcomes or mechanically impute returns.

Raw returns range from −92.2% to 3,017.6%. We cap observations at each month's linearly interpolated P1 and P99; 2.02% are changed (Figure A4). The P/B score has an unusual 0.50 spike, associated with low book-equity ranks. A plausible explanation is a neutral replacement score for observations whose valuation ratio is unavailable or economically problematic, potentially including nonpositive book equity. French's HML construction requires positive book equity [1], so the anomaly matters for a comparable value sort. The ranks cannot establish either raw equity signs or the vendor's replacement rule. We therefore exclude these stock-months from baseline value portfolios, retain them in the market proxy, and test their inclusion separately (Figure A5).

To assess P/B = market capitalization / book equity with percentile inputs, we hold one characteristic approximately constant using ten score bands within each month. Spearman correlations of the other two should be positive within book-equity bands, negative within market-cap bands, and positive within P/B bands, assuming positive equity. The corresponding median correlations are 0.898, −0.897 and 0.977 after excluding Pb = 0.50 (Figure A6; Table A5). These signs support directional consistency. Dividing percentile

scores would not reconstruct the accounting ratio, because ranks discard the original distances between values.

2B Building a market reference

We normalize supplied market-cap scores into long-only weights. Doubling a percentile score does not imply twice the actual capitalization. December 1998–November 2018 formation data map to January 1999–December 2018 returns. The proxy and MSCI USA have monthly correlation 0.907 and matching signs in 89.6% of months (Figure A7; Table A6). Their terminal wealth differs substantially, so the comparison supports broad market coverage without validating every stock or equating rank-based weights with true capitalization.

2C Comparing three value portfolio rules

We reverse the supplied P/B percentile score as B/P score = 1.01 − Pb, without treating the score as a raw valuation ratio. Each month, after excluding Pb = 0.50, we construct three fully invested portfolios: equal weights in the cheapest 20%, including boundary ties; weights proportional to B/P scores; and weights proportional to B/P scores times market-cap scores. Returns use the same monthly winsorization as the market proxy, which retains Pb = 0.50. All strategies rebalance monthly without transaction costs.

Over 1999–2018, the three value strategies have CAGRs of 17.15%, 13.16% and 9.92%, versus 8.77% for the market proxy (Figure A8). The top-20% strategy also has the highest annualized volatility, 22.02%, compared with 18.70%, 17.32% and 16.48%. The top-20% strategy invests only in the cheapest stocks, whereas the other two strategies also allocate weight to relatively more expensive stocks, diluting their value exposure. Equal weighting avoids giving larger stocks additional weight within the selected group. These features may help explain its higher return and volatility, although this comparison cannot separate the effects of stock selection and weighting. Retaining Pb = 0.50 produces CAGRs of 17.05%, 13.32% and 10.02%, preserving the ordering.

At the final formation date, November 2018, the top-20% portfolio holds 218 stocks and has the largest single position, 0.4587%. The other two hold 1,057 stocks, with maximum weights of 0.1849% and 0.3954%. Equal allocation across a much narrower selected set explains its relatively large maximum weight; multiplying value and size scores also increases concentration compared with value-score weighting alone (Table A7).

2D Active returns and similarity to HML

We subtract the market proxy from each strategy each month. Tracking errors are 9.51%, 4.94% and 2.86%, respectively (Table A7). Retaining size-score weighting makes the value-times-size portfolio structurally closest to its benchmark. Active-return correlations across the value strategies are 0.740–0.957. Correlations with HML are 0.555, 0.598 and 0.855 (Table A8), supporting a common value theme with meaningful implementation differences. The stronger HML match for value-times-size is consistent with capitalization weighting within French’s size/value sorts, although our strategy does not balance separate small and big spreads [1].

Sources: instructor data and cited methods. Linked results: project repository

Part 3 Industry portfolio optimization

3A Estimating low volatility industry allocations

We next allocate across 49 value-weighted industry portfolios using January 1995–December 2004 total returns. Missing-return sentinels are checked before percent-to-decimal conversion; all 120 estimation months are complete. With sample covariance Σ, the long-only problem is:

$$\min_w \; w^T \Sigma w \quad \text{subject to} \quad 1^T w = 1,\; w_i \geq 0$$

We solve the long-only problem in Python using scipy.optimize.minimize with the SLSQP algorithm. Starting from equal weights, we minimize annualized variance, $12 w^T \Sigma w$, with analytical gradient $24 \Sigma w$, nonnegative weight bounds and the equality constraint $\Sigma_i w_i = 1$. The stopping tolerance is 10^{-12}, with at most 5,000 iterations. We require the solver to report successful convergence, then check that weights sum to one and satisfy nonnegativity within a numerical tolerance of 10^{-8}. Multiplying variance by 12 changes its units but not the optimal weights.

For the unrestricted minimum-variance portfolio, NumPy solves $\Sigma z = 1$ and we normalize z by its sum, allowing negative weights. The inverse-volatility allocation normalizes $1/\sigma$ across industries and does not directly use correlations. All three allocations pass full-investment checks; their fitted variances also satisfy unrestricted ≤ long-only ≤ inverse-volatility. Table A9 reports every weight, totals and the five highest/lowest weights. Listing A1 provides the optimization code.

3B and 3C In sample and out of sample risk

The unrestricted minimum-variance, long-only minimum-variance and inverse-volatility portfolios have 1995–2004 annualized volatilities of 7.59%, 10.32% and 13.99% (Table A10). This ordering follows the feasible sets: allowing shorts relaxes constraints, while inverse-volatility weights are a feasible long-only allocation. The in-sample comparison measures how well each rule fits the estimated covariance structure.

Holding the 2004 target weights fixed and rebalancing monthly, volatilities for January 2005– July 2023 become 16.75%, 14.29% and 17.23%. Long-only optimization is now the least volatile. Inverse-volatility weighting has the highest realized risk in both windows: it reduces individual volatility exposure but does not optimize diversification through correlations. All calculations use sample monthly standard deviation times √12 and exclude costs.

The reversal illustrates why the sample covariance optimum need not minimize future risk. Only 120 monthly observations estimate dependence among 49 industries. An unrestricted optimizer can exploit small estimated differences through offsetting long and short positions; errors or changing correlations can then materially change realized risk. Long-only constraints restrict these positions and can improve robustness despite raising the fitted minimum variance. The result supports constraints in this sample, without implying that long-only optimization must win in every future period (Table A10).

Sources: instructor data and cited methods. Linked results: project repository

3D Updating the covariance estimates

We re-estimate a long-only allocation at December 2015 from January 2006–December 2015 returns. Over the shared January 2016–July 2023 window, the old and updated allocations have volatilities of 14.52% and 12.81%, a reduction of 1.71 percentage points (Table A10). The identical evaluation window makes this a comparison of allocation rules under the same realized market conditions. The newer estimation period includes the financial crisis and may better reflect later industry relationships. Nevertheless, this single outcome cannot identify why the update helped: fresh estimates also contain noise, and future regime changes or turnover costs may offset their benefit.

3E Crisis performance and industry resilience

Figure A9 compounds $1 from December 31, 2004 using the original long-only allocation and MSCI USA, with a base-two logarithmic axis. During September 2008–February 2009, the strategy loses 40.33% versus 41.76% for USA, outperforming by 1.44 percentage points. During February–March 2020, it loses 21.25% versus 19.80%, underperforming by 1.44 points (Table A11). Minimizing historical variance does not guarantee smaller losses in every crisis.

The financial-crisis top ten include food, pharmaceuticals, beverages, tobacco and utilities, broadly consistent with less cyclical demand. Precious metals rank first, while personal services, retail and restaurants/hotels are less obvious defensive labels (Table A12). Industry aggregates mix different businesses, so demand stability alone cannot explain each ranking. Gold denotes precious-metal equities, not a bullion holding.

The COVID top ten retain pharmaceuticals, food, tobacco and consumer goods, but also include software, retail, measuring/control equipment, shipping containers and business supplies. Software and parts of retail plausibly benefited from remote work and online activity; OECD documents the pandemic's acceleration of these activities [5]. That interpretation does not establish a causal explanation for our two-month ranking, especially for broad categories such as retail.

Crisis resilience is also distinct from historically low volatility. In the original 1995–2004 estimation window, precious metals and software have annualized volatilities of 42.5% and 33.2%, both above the 49-industry median of 21.9%. All ten leading industries still lose money in each crisis window. Their rankings show relative resilience, not universally positive pandemic returns or an allocation identifiable in advance.

Part 4 Product innovation

4A and 4B Risk and the daily leverage objective

We evaluate 147 paired daily total returns for NVDA and NVDL from February 1–August 30, 2024. Daily sample standard deviations are 3.746% and 7.479%; 95% historical VaRs are 6.121% and 12.029%; and CVaRs are 7.073% and 14.162% (Table A13). VaR uses the negative interpolated 5th return percentile; CVaR averages the eight observations at or below it. These daily estimates show that leverage increases both volatility and the size of extreme losses.

OLS of NVDL on NVDA with an intercept gives a slope of 1.995975 and 95% interval [1.987407, 2.004543]. The interval includes two, consistent with the average daily target specified in the assignment (Figure A10). It does not establish exact delivery on every day or remove the distinction between a pre-expense objective and realized returns. Conventional standard errors use 145 residual degrees of freedom.

Sources: instructor data and cited methods. Linked results: project repository

4C Costs and unexplained performance drag

The daily intercept (Table A13) is −0.00061164, with interval [−0.00093462, −0.00028867]. Multiplication by 252 implies approximately −15.41% annualized drag, much larger than the assigned historical 1.15% net expense ratio [6]. Financing or swap costs and trading spreads or market impact may contribute. This fitted intercept is a crude conditional estimate; it neither measures the fund's annual fee directly nor identifies each cost. The provided total-return convention has not been reconstructed as NAV versus market-price returns.

4D and 4E Multi day compounding

Over February–August, NVDA returns 94.05% and NVDL 181.53%, versus 188.10% for twice NVDA's cumulative gain. The full-period shortfall is 6.57 percentage points, while monthly gaps have both signs. In August, NVDA gains 2.01% while NVDL loses 1.60% (Table A14; Figure A11). For the hypothetical +10%, −9% stock path, the stock returns $1.10 \times 0.91 - 1 = +0.10\%$, while an ideal daily 2× fund returns $1.20 \times 0.82 - 1 = -1.60\%$. Daily reset and compounding make a multi-day outcome path dependent, even without expenses.

4F Matching the product to the holding period

For the hypothetical bullish announcement-day trader who deliberately seeks larger exposure and can bear larger losses, NVDL aligns with a daily 2× objective. For an investor expecting a volatile but favorable year, NVDA is the more straightforward exposure: the ETF does not promise twice the annual return. Spreads, financing and tracking also matter. A bullish view alone does not remove either product's downside.

Part 5 Portfolio analytics

5A Comparing definitions of value and growth

MSCI's February 2021 method uses book/price, forward earnings/price and dividend yield for value; growth uses long-and short-term forward EPS growth, internal growth, historical EPS growth and historical sales-per-share growth [3]. Value and growth are scored separately; securities can be allocated partly to both style indexes. Long-term forward growth is excluded for Small Cap. Sales growth is excluded for Banks and Diversified Financials, apart from specified Multi-Sector Holdings and Financial Exchanges & Data sub-industries. Missing inputs are excluded; Standard indexes double-weight long-term forward growth. HML averages high-minus-low book-to-market returns across small and big capitalization groups, whereas MSCI Value is long only. These differences motivate exposure and alpha analysis; the 2021 methodology is not assumed unchanged throughout the historical sample.

5B Estimated exposures of MSCI Value

We use the same 343 monthly observations as Part 1. The dependent variable is MSCI USA Standard Value Gross USD total return; the regressors are an intercept, RF, Mkt−RF, HML and SMB. This is a monthly time-series regression of one index, although the question calls it a pooled panel. RF enters separately because Mkt−RF, HML and SMB are already return spreads. Its coefficient is estimated rather than fixed at one. Conventional OLS t statistics are reported without Newey–West adjustment.

$$\text{Value}_t = \alpha + b_{RF}\,\text{RF}_t + b_M\,(\text{Mkt}-\text{RF})_t + b_H\,\text{HML}_t + b_S\,\text{SMB}_t + \varepsilon_t$$

The market coefficient is 0.9480 (t = 76.99), HML is 0.3216 (t = 19.68), and SMB is −0.1744 (t = −9.90). RF is 0.8646 (t = 2.79). R^2 is 0.9484 (Table A15). Positive value exposure and negative SMB exposure are consistent with the advertised value and large/mid-cap

Sources: instructor data and cited methods. Linked results: project repository

characteristics. The RF t statistic tests zero, not one; each loading summarizes an average sample relationship rather than identical stock selection.

5C Additional performance beyond factor exposure

The intercept is −0.0713% per month, approximately −0.855% per year using arithmetic annualization, with t = −0.906 and p = 0.366 (Table A15). We find no statistically significant positive alpha for MSCI Value relative to this factor specification. A richer value definition therefore does not show a detectable return improvement after fitted factor exposure in this sample. The exercise does not isolate its causal contribution, because universe, weights and other omitted exposures also differ.

5D Constructing and evaluating the replication

We form replication returns from the estimated factor coefficients, excluding the intercept:

$$\text{Replication} = 0.8646\ \text{RF} + 0.9480\ (\text{Mkt}-\text{RF}) + 0.3216\ \text{HML} - 0.1744\ \text{SMB}$$

The correlation with Value is 0.9739. Value has 9.87% annualized arithmetic return, 15.27% volatility and a 0.505 Sharpe; replication has 10.73%, 14.87% and 0.576 (Figure A12; Table A16). Replication is effective at matching monthly variation and is historically preferable on these gross sample metrics. Omitting a negative fitted intercept mechanically increases its mean return by approximately 0.855 percentage points per year. Because estimation and evaluation use the same observations, this is not independent evidence of future superiority.

Three implementation questions remain before choosing an investment. First, management, financing, borrowing and transaction costs can reverse a gross advantage. Second, the long–short factor legs, including negative SMB exposure, require feasible liquid instruments and practical rebalancing; the coefficients are factor exposures rather than ordinary long-only fund weights. Third, exposures and performance need validation in a later period. High correlation alone does not guarantee identical cumulative wealth or small tracking error.

These results extend the construction comparisons in Parts 1 and 2. A strategy can retain a recognizable factor identity without being the most efficient implementation of that identity. We next examine the multifactor allocation relative to its market benchmark, where the relevant denominator is active risk rather than total volatility.

5E Benchmark relative performance

For each component and the monthly 25%-weighted multifactor allocation, active return is its return minus MSCI USA. Over January 1995–July 2023, the multifactor allocation earns 0.152% annualized arithmetic active return with 3.387% tracking error. Its information ratio is 0.045 (Table A17). The excess return is small relative to active-return variability, despite the Sharpe improvement documented in Part 1. Sharpe and information ratios answer different questions because one uses a risk-free comparison and total risk, while the other uses an equity benchmark and active risk.

Each individual component has higher tracking error than the mix, ranging from 5.154% to 7.730%. Imperfect co-movement therefore diversifies active risk as well as total risk. Value and Minimum Volatility underperform USA on arithmetic mean returns in this period; Momentum contributes the strongest positive active mean. The combined result is historical, gross of costs and close to the benchmark in average return.

Sources: instructor data and cited methods. Linked results: project repository

5F Decomposing tracking error

Let A denote multifactor active return and A_i each component's active return. Since A is their equal-weighted average, $\mathrm{Var}(A) = \frac{1}{4} \sum_i \mathrm{Cov}(A, A_i)$. Consistent sample covariances allocate active variance as 23.26% to Value, 34.08% to Minimum Volatility, 17.86% to Equal Weight and 24.81% to Momentum (Figure A13; Table A18). These are covariance contributions, not each component's standalone variance. The corresponding allocations of annualized tracking error sum to 3.387%.

Minimum Volatility is the largest relative-risk contributor even though its total volatility is low. Absolute risk measures fluctuations in portfolio returns; tracking error measures fluctuations in the difference from the benchmark. A defensive allocation can therefore have low total risk and substantial active risk when it responds differently to market movements. In a strong rally, that difference can produce sustained benchmark underperformance despite fulfilling an absolute-risk objective.

This distinction connects the result to limits to arbitrage. Managers judged against an equity benchmark may face client withdrawals or career pressure before a defensive position pays off, discouraging them from maintaining it. Investors unable to borrow may instead seek higher exposure through high-beta stocks, supporting demand for those stocks and limiting arbitrage of low-risk premia [4]. Minimum Volatility may therefore suit an investor who wants lower overall risk and accepts that it can lag the market. For an investor judged against MSCI USA, its 34.08% share of active risk needs more attention. The choice depends on the investor's objective; the strategy is not always the better option (Figure A13; Table A18).

Overall assessment

Factor labels remain useful, but construction, estimation and evaluation choices materially affect the investment outcome. The sample favors constraints over unrestricted optimization out of sample, shows the holding-period limits of daily leverage, and finds little multifactor excess return relative to active risk. Implementation decisions should account for these distinctions and for costs.

References

[1] Kenneth R. French Data Library. Three-factor, momentum-decile, beta-decile and 49-industry descriptions; classroom historical return files.

[2] MSCI. Momentum Indexes Methodology, December 2013, pp. 4–10.

[3] MSCI. Global Investable Market Value and Growth Index Methodology, February 2021, pp. 5, 9–13.

[4] Frazzini, A., and Pedersen, L. H. (2014). Betting Against Beta. Journal of Financial Economics, 111, 1–25.

[5] OECD (2020). OECD Digital Economy Outlook 2020, executive summary.

[6] FE5108 Fall 2026 complete project handout and instructor-provided data. NVDL objective and historical 1.15% expense ratio are as stated in the handout.

Sources: instructor data and cited methods. Linked results: project repository

Appendix Supporting exhibits and code

Part 1 Factor investing

Table A1. Mean factor returns and t statistics before and after 1992.
Cited in main text: p. 1, section 1A.

Period	Factor	Months	Monthly mean	Annual mean	Mean t
1927–1992	Mkt-RF	792	0.655%	7.86%	3.219
1927–1992	SMB	792	0.240%	2.88%	2.126
1927–1992	HML	792	0.435%	5.22%	3.333
1993–Jul 2023	Mkt-RF	367	0.719%	8.62%	3.091
1993–Jul 2023	SMB	367	0.099%	1.19%	0.599
1993–Jul 2023	HML	367	0.146%	1.75%	0.830

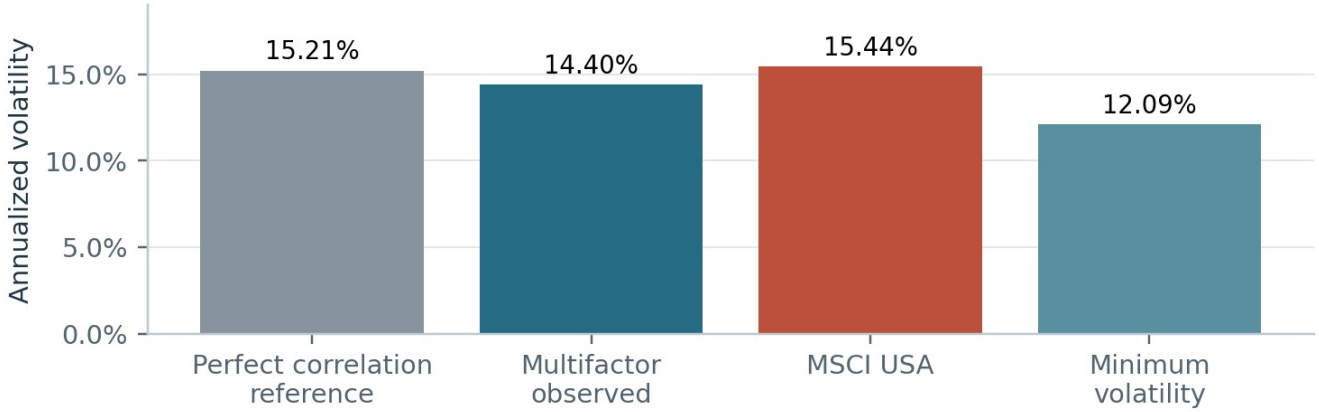
Source: classroom US French three-factor data [1]. January 1927–December 1992 and January 1993–July 2023. Arithmetic means; ordinary mean t statistics without Newey–West adjustment.

Table A2. Annualized returns, volatility and Sharpe ratios of MSCI indexes and RF.
Cited in main text: p. 1, section 1C.

Index or asset	Annual mean	Annual volatility	Sharpe
USA	11.22%	15.44%	0.586
Value	9.87%	15.27%	0.505
Minimum volatility	10.42%	12.09%	0.683
Equal weight	11.81%	17.13%	0.563
Momentum	13.38%	16.34%	0.686
Multifactor	11.37%	14.40%	0.639
RF	2.16%	0.61%	0.000

Source: classroom MSCI USA Standard large/mid-cap Gross USD levels and French RF. January 1995–July 2023, n = 343. Mean = 12 × monthly arithmetic mean; volatility = √12 × sample standard deviation. RF's zero Sharpe follows the literal formula and is not an informative ranking.

Figure A1. Multifactor volatility and the perfect-correlation reference.
Cited in main text: p. 1, section 1C.



Same source and period as Table A2. The first bar holds component volatilities fixed and sets every pairwise correlation to +1; it is an analytical reference, not an investable portfolio. Values are rounded to match the displayed table.

Table A3. Correlations between MSCI active returns and academic factor spreads.
Cited in main text: p. 2, section 1D.

MSCI less USA	HML	SMB	Momentum	Low beta
Value	0.728	-0.177	-0.243	0.415
Minimum volatility	0.232	-0.307	0.303	0.815
Equal weight	0.431	0.322	-0.448	-0.199
Momentum	-0.223	0.079	0.643	0.140

Source: classroom MSCI Gross USD levels and French value-weighted research portfolios. January 1995–July 2023, n = 343 in every cell. Momentum is highest minus lowest momentum decile; low beta is lowest minus highest beta decile.

Figure A2. Momentum returns and trailing twelve-month market volatility.
Cited in main text: p. 2, section 1E.

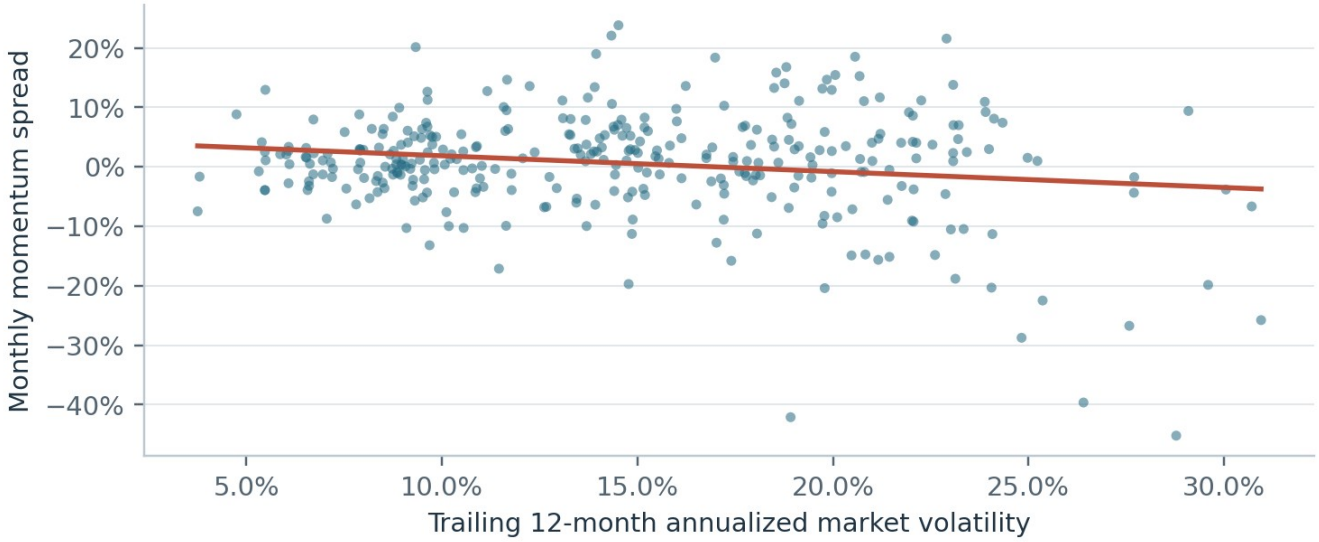


Table A4. Momentum regressions and mean returns across volatility regimes.
Cited in main text: p. 2, section 1E.

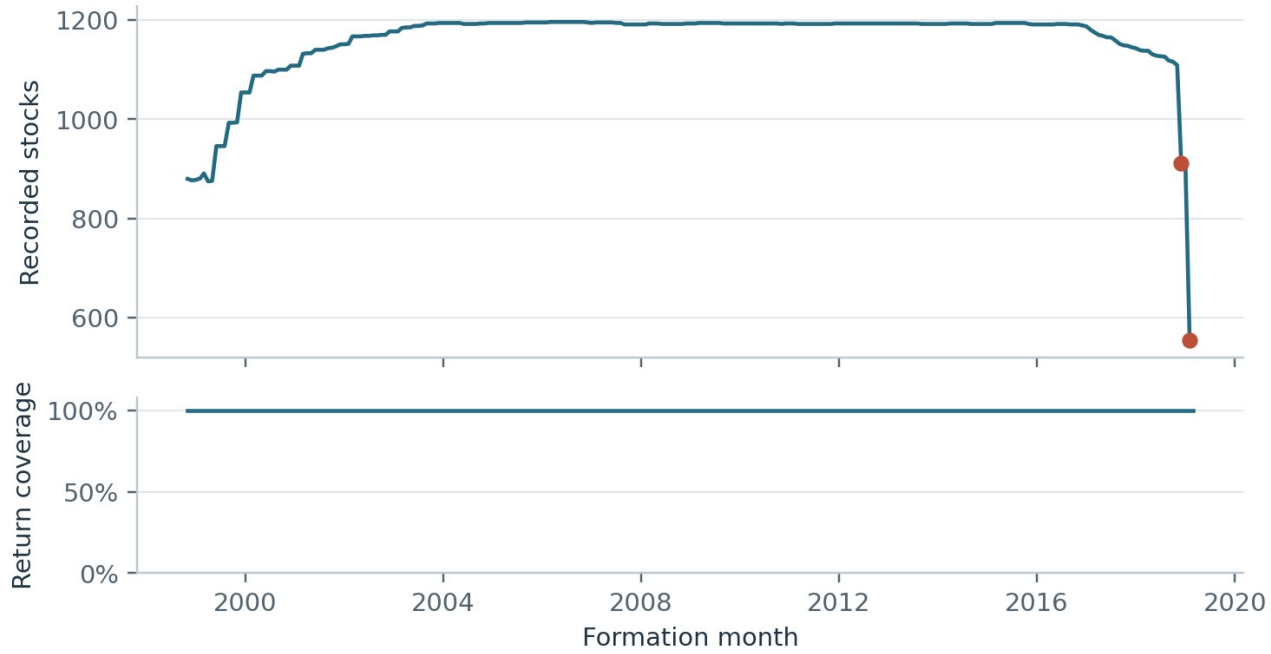
Volatility window	Months	Slope	HAC SE	p value	R^2
Current month	332	-0.267	0.116	0.022	3.2%
Prior month	332	-0.221	0.106	0.036	2.2%

Current volatility group	Months	Annual mean momentum
Low	111	12.47%
Middle	110	22.34%
High	111	-12.33%

Source: classroom US Mkt−RF and momentum deciles, December 1995–July 2023, n = 332. Predictor is trailing twelve-month sample volatility × √12. Bartlett HAC(12), T/(T−2) correction and asymptotic normal two-sided p values. Terciles use full-sample ranks; they are descriptive rather than real-time thresholds.

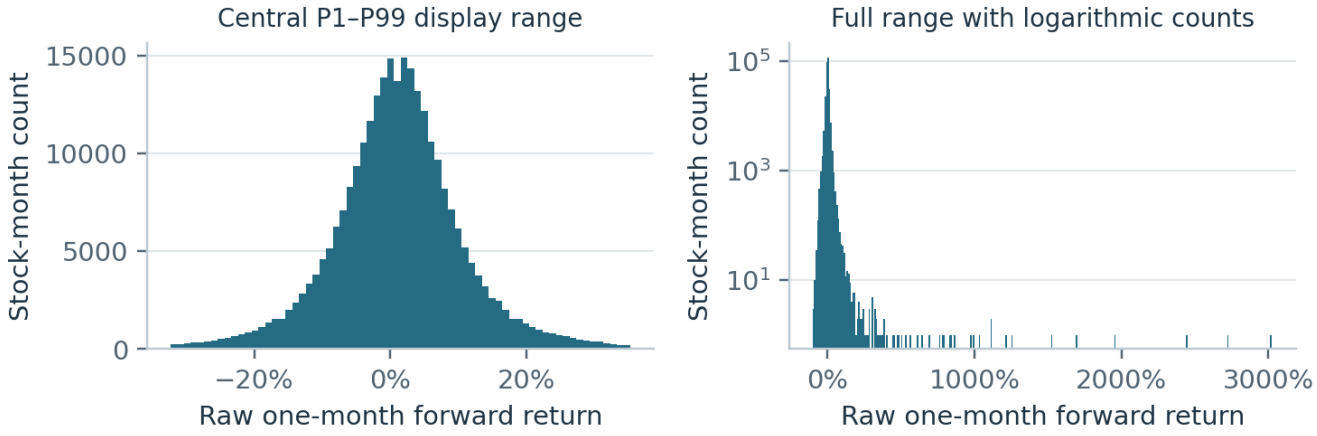
Part 2 Security level portfolio construction

Figure A3. Monthly stock counts and coverage of the return field.
Cited in main text: p. 2, section 2A.



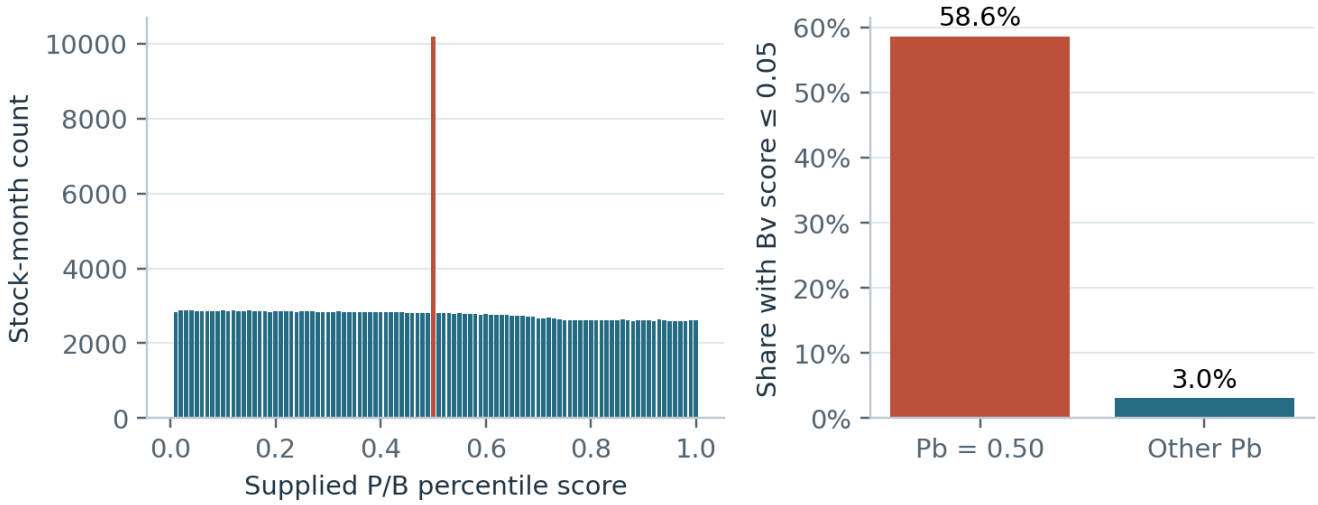
Source: instructor security panel. All 245 formation months, November 1998–March 2019; 283,380 stock-months. The highlighted count declines occur in December 2018 and February 2019. Zero returns are observed values. Coverage is conditional on stocks recorded in each month.

Figure A4. Raw forward-return distributions, central and full ranges.
Cited in main text: p. 2, section 2A.



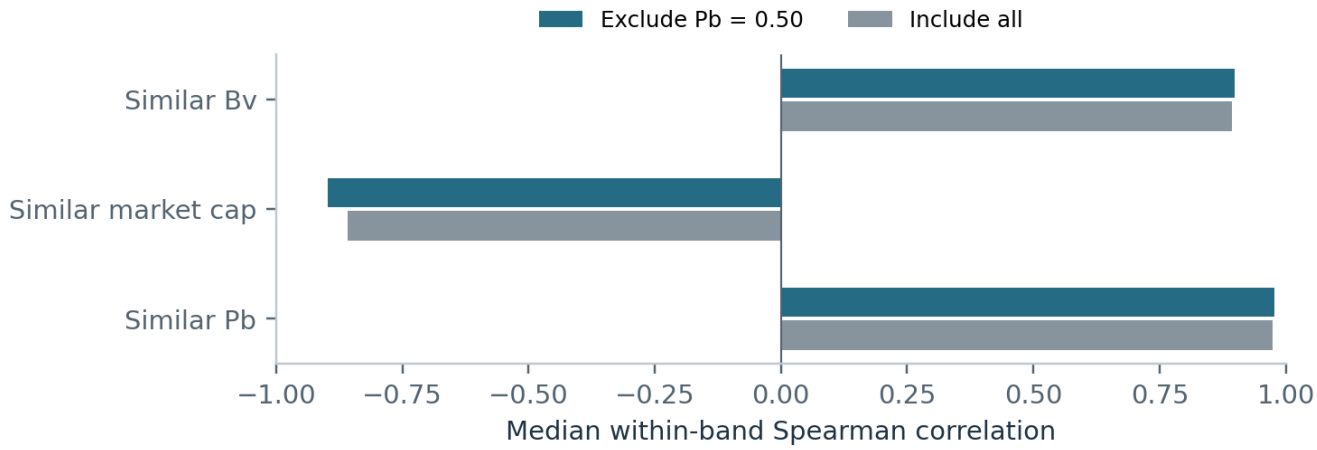
Same panel and formation period as Figure A3. Returns are decimals and one month forward. The central display excludes pooled P1/P99 tails only for visualization. Cleaning instead caps returns at P1/P99 separately within each formation month, with linear interpolation. The full-range horizontal axis is linear.

Figure A5. P/B-score frequencies and the association of Pb = 0.50 with low book equity.
Cited in main text: p. 2, section 2A.



Source: instructor security panel, November 1998–March 2019. The anomalous group has 10,197 observations, 3.60% of the sample. Bv ≤ 0.05 occurs in 58.6% of this group versus 3.0% elsewhere. The rank scale does not identify the sign of raw book equity.

Figure A6. Conditional rank correlations for directional valuation checks.
Cited in main text: p. 2, section 2A.



Source: instructor US stock panel, November 1998–March 2019; monthly score bands. Spearman methods and medians appear in Table A5.

Table A5. Conditional valuation checks excluding Pb = 0.50.
Cited in main text: p. 2, section 2A.

Approximately held constant	Expected sign	Median rho excluding 0.50	Groups with expected sign
Book equity	+	0.898	100.0%
Market cap	–	-0.897	100.0%
P/B	+	0.977	100.0%

Same source and period. Ten fixed score bands per month; within-band Spearman correlations use average ranks for ties. Summary values are medians across valid month-band groups. Groups below ten observations or with constant compared values are omitted and counted in the source tables. This is a directional check, not reconstruction of raw accounting ratios.

Figure A7. Market-proxy wealth and monthly-return comparison with MSCI USA.
Cited in main text: p. 3, section 2B.

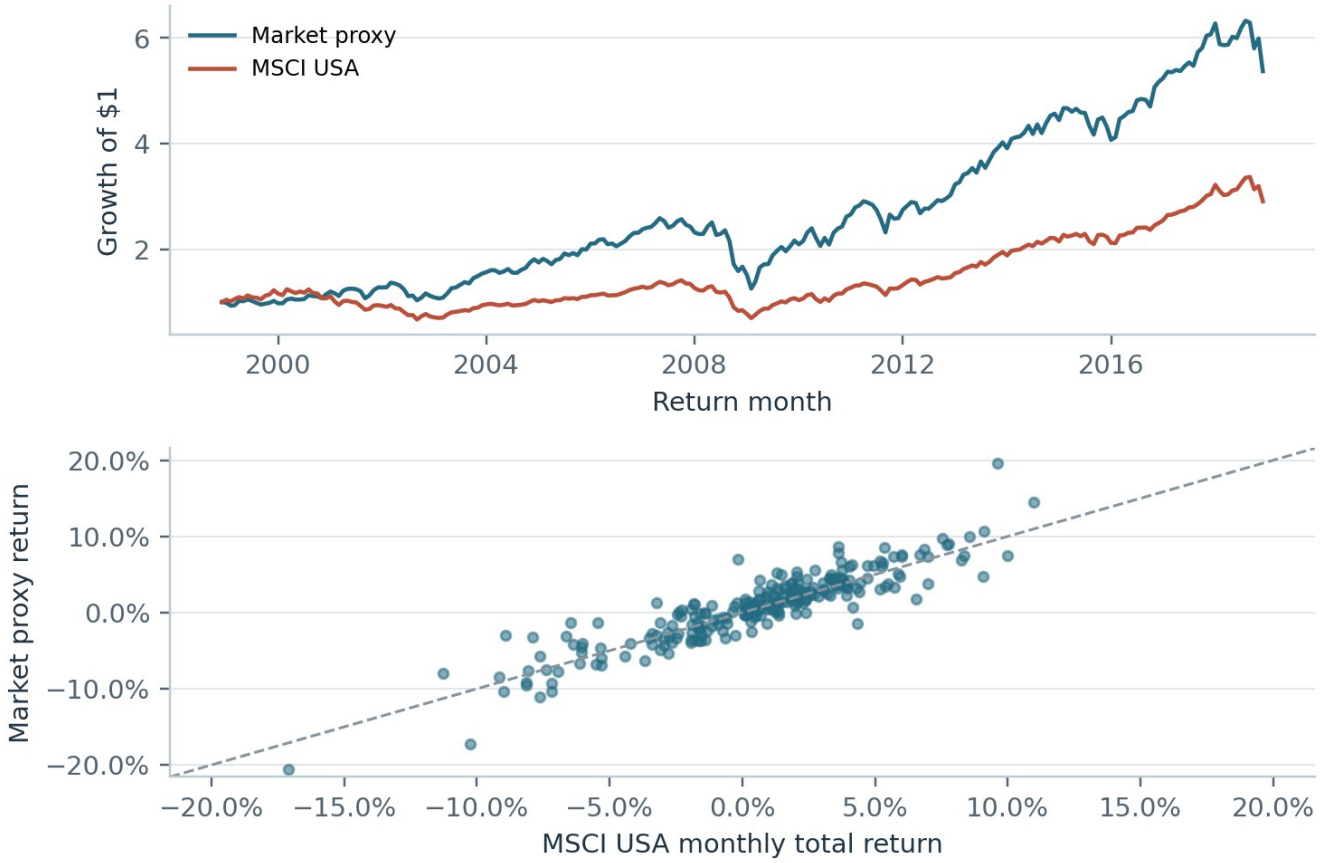


Table A6. Market-proxy and MSCI USA comparison metrics.
Cited in main text: p. 3, section 2B.

Comparison metric	Result
Monthly return correlation	0.907
Same-sign months	89.6%
Terminal proxy wealth	$5.37
Terminal MSCI USA wealth	$2.91

Source: instructor stock panel and shared MSCI Gross USD workbook. Return window: January 1999–December 2018, n = 240. Proxy weights normalize supplied market-cap scores and use monthly winsorized forward returns. MSCI levels are converted to same-month total returns; the dotted scatter line is y = x. Both wealth paths begin at $1 on December 31, 1998; displayed MSCI wealth includes that baseline.

Figure A8. Cumulative wealth of three value portfolios and the market proxy.
Cited in main text: p. 3, section 2C.

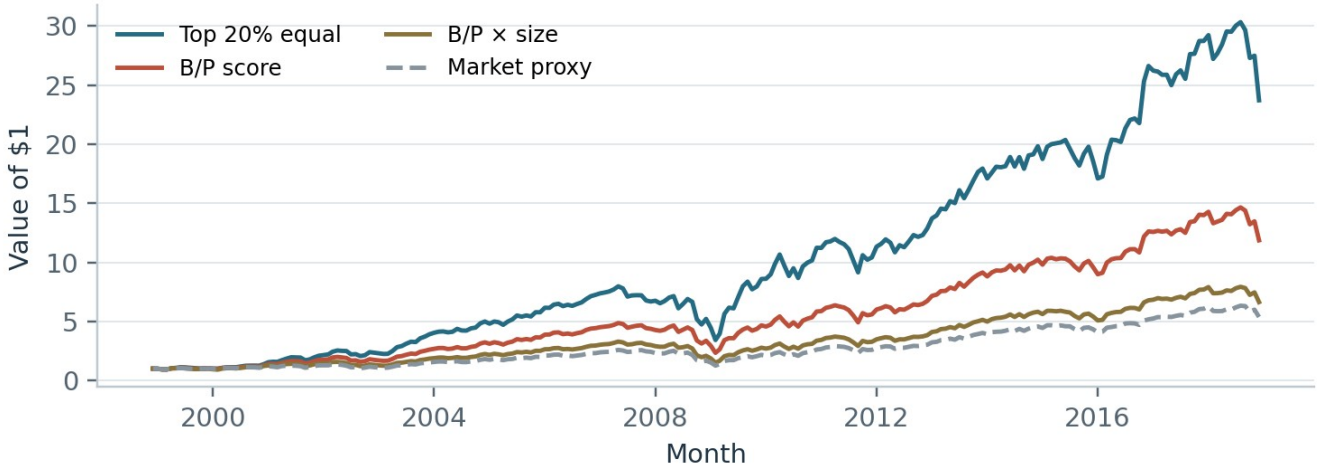


Table A7. Value portfolio performance, tracking error and final-period holdings.
Cited in main text: p. 3, section 2C and 2D.

Strategy	CAGR	Volatility	TE	Stocks	Max weight
Top 20% equal	17.15%	22.02%	9.51%	218	0.4587%
B/P score	13.16%	18.70%	4.94%	1057	0.1849%
B/P × size	9.92%	17.32%	2.86%	1057	0.3954%
Market proxy	8.77%	16.48%	0.00%	–	–

Source: instructor stock panel. January 1999–December 2018, n = 240. All portfolios rebalance monthly using P1/P99-cleaned returns. Value strategies exclude Pb = 0.50; market retains it. CAGR uses geometric compounding; volatility/TE use sample monthly standard deviation × √12. Holdings refer to November 2018 formation weights for December returns. Market holdings are outside the requested value-holdings comparison.

Table A8. Correlations among value active returns and HML.
Cited in main text: p. 3, section 2D.

Active series	Top 20%	B/P score	B/P × size	HML
Top 20% equal	1.000	0.957	0.740	0.555
B/P score	0.957	1.000	0.764	0.598
B/P × size	0.740	0.764	1.000	0.855
HML	0.555	0.598	0.855	1.000

Source: same stock panel and classroom HML, January 1999–December 2018, 240 aligned months. Each strategy subtracts the score-weighted market proxy. HML is already a long–short spread and is neither winsorized nor reduced by the market again.

Part 3 Industry portfolio optimization

Table A9. Industry weights; green marks the five highest and red the five lowest in each strategy.
Cited in main text: p. 4, section 3A.

Industry code (Table A9)	Minimum variance shorts allowed	Minimum variance long only	Inverse volatility
Agric	6.38%	10.68%	2.23%
Food	19.48%	9.05%	2.91%
Soda	-5.14%	0.00%	1.46%
Beer	-1.81%	0.00%	2.11%
Smoke	0.07%	0.00%	1.45%
Toys	-0.52%	0.00%	2.05%
Fun	1.32%	0.00%	1.77%
Books	41.28%	7.72%	2.69%
Hshld	21.15%	12.28%	2.75%
Clths	2.26%	0.00%	2.00%
Hlth	-4.92%	0.00%	1.75%
MedEq	25.20%	5.86%	2.86%
Drugs	-1.18%	9.48%	2.52%
Chems	-28.41%	0.00%	2.41%
Rubbr	9.44%	0.00%	2.22%
Txtls	0.08%	0.00%	2.09%

Industry code (Table A9)	Minimum variance shorts allowed	Minimum variance long only	Inverse volatility
BldMt	-3.76%	0.00%	2.42%
Cnstr	-17.80%	0.00%	1.87%
Steel	6.85%	0.00%	1.45%
FabPr	-2.40%	0.00%	1.67%
Mach	-2.33%	0.00%	1.82%
ElcEq	-8.14%	0.00%	1.98%
Autos	6.33%	0.00%	1.81%
Aero	-20.62%	0.00%	1.82%
Ships	-0.68%	0.00%	1.94%
Guns	13.66%	9.30%	1.71%
Gold	-1.12%	0.73%	1.04%
Mines	5.22%	0.00%	2.00%
Coal	-1.73%	0.00%	1.00%
Oil	19.27%	1.60%	2.52%
Util	21.51%	16.73%	2.81%
Telcm	0.15%	2.89%	2.02%
PerSv	-13.47%	0.00%	2.15%
BusSv	-32.33%	0.00%	2.40%
Hardw	1.97%	0.00%	1.20%
Softw	10.29%	0.00%	1.33%
Chips	1.33%	0.00%	1.17%
LabEq	-4.16%	0.00%	1.58%
Paper	25.99%	0.00%	2.47%
Boxes	-3.30%	0.00%	1.86%
Trans	7.77%	0.00%	2.44%
Whlsl	-5.28%	0.00%	2.60%
Rtail	0.60%	0.00%	2.28%
Meals	13.86%	0.00%	2.47%
Banks	-16.55%	0.00%	2.26%
Insur	-3.99%	0.00%	2.45%
RlEst	23.00%	13.67%	2.79%
Fin	-4.47%	0.00%	1.61%
Other	-0.31%	0.00%	1.80%
Total	**100.00%**	**100.00%**	**100.00%**

Source: instructor French 49-industry value-weighted total returns, January 1995–December 2004, 120 months. Weights sum to one before rounding; displayed zero weights can include tiny numerical values. Exact ties use original industry order, so only five of multiple zero weights are colored red. No transaction costs.

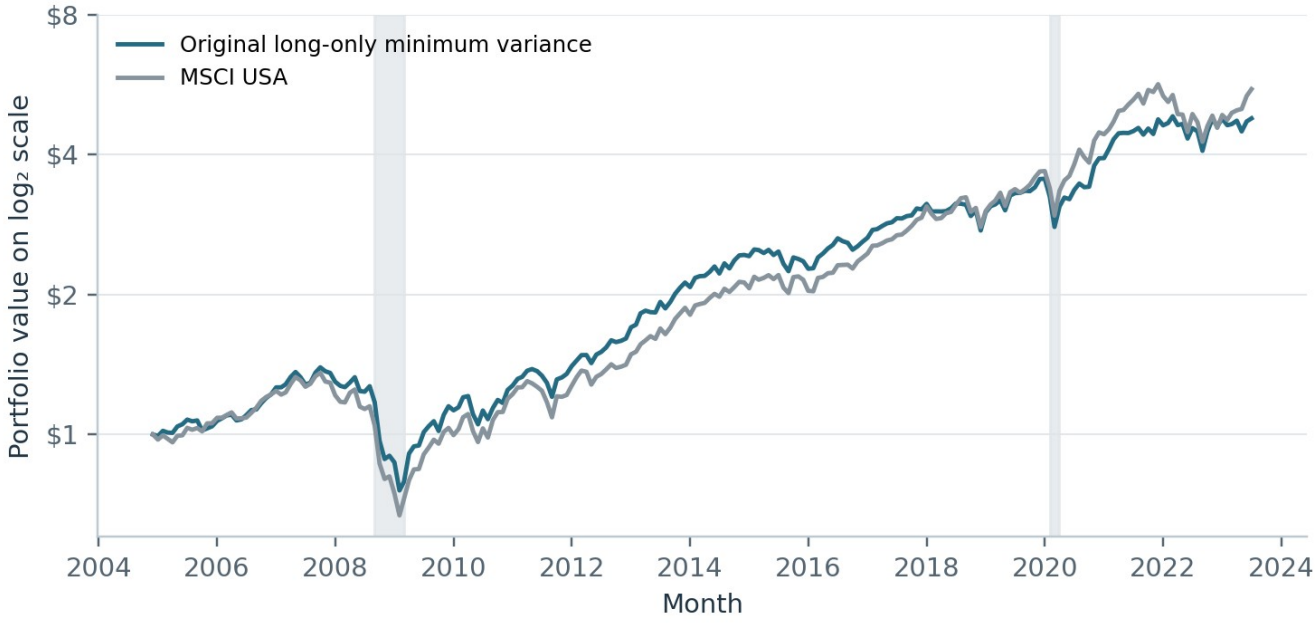
Table A10. Annualized volatility in the estimation, evaluation and reoptimization windows.
Cited in main text: pp. 4–5, section 3B–3D.

Portfolio	1995–2004 in sample	2005–Jul 2023 out of sample
Minimum variance with shorts	7.59%	16.75%
Minimum variance long only	10.32%	14.29%
Inverse volatility	13.99%	17.23%

Long-only allocation	Estimation window	2016–Jul 2023 volatility
Original	1995–2004	14.52%
Updated	2006–2015	12.81%

Source: same 49-industry panel. Evaluation counts are 120, 223 and 91 months. Annualized volatility = √12 × monthly sample standard deviation. Fixed target weights are rebalanced monthly. Updated allocation reduces later-window volatility by 1.71 percentage points.

Figure A9. Wealth of the original long-only portfolio and MSCI USA on a log-2 scale.
Cited in main text: p. 5, section 3E.



Source: instructor French industry returns and MSCI USA Standard Gross USD levels. $1 starts December 31, 2004; January 2005–July 2023 returns follow. Vertical axis uses log base two. Shading marks September 2008–February 2009 and February–March 2020. The strategy uses the original 2004 weights, with monthly rebalancing and no costs.

Table A11. Compounded crisis returns of the original long-only portfolio and MSCI USA.
Cited in main text: p. 5, section 3E.

Crisis	Long-only portfolio	MSCI USA	Difference pp
Sep 2008–Feb 2009	-40.33%	-41.76%	1.44
Feb–Mar 2020	-21.25%	-19.80%	-1.44

Source: instructor 49-industry returns and MSCI USA Gross USD levels; six and two months respectively. Returns are products of (1 + monthly return) minus one. Differences use unrounded returns.

Table A12. Ten most resilient industries in each crisis; all returns are negative.
Cited in main text: p. 5, section 3E.

Rank	Financial crisis Sep 2008–Feb 2009	Return	COVID Feb–Mar 2020	Return
1	Precious metals (Gold)	-11.64%	Precious metals (Gold)	-3.21%
2	Beer and liquor (Beer)	-22.26%	Pharmaceuticals (Drugs)	-7.07%
3	Personal services (PerSv)	-24.08%	Retail (Rtail)	-10.26%
4	Pharmaceuticals (Drugs)	-24.14%	Tobacco products (Smoke)	-12.71%
5	Food products (Food)	-24.32%	Food products (Food)	-13.14%
6	Restaurants and hotels (Meals)	-26.06%	Computer software (Softw)	-14.57%
7	Candy and soda (Soda)	-29.43%	Consumer goods (Hshld)	-14.64%
8	Tobacco products (Smoke)	-29.61%	Measuring and control equipment (LabEq)	-15.88%
9	Retail (Rtail)	-30.02%	Shipping containers (Boxes)	-16.23%
10	Utilities (Util)	-32.58%	Business supplies (Paper)	-16.47%

Historical volatility context	1995–2004 annualized volatility
Precious metals (Gold)	42.48%
Computer software (Softw)	33.21%
Median across 49 industries	21.92%

Source: instructor value-weighted French 49-industry monthly total returns. Rankings use the separate crisis windows, not post-crisis recovery months. The volatility context uses the original estimation window and sample standard deviations. Gold is a precious-metal equity industry; LabEq is broader than medical equipment, and Paper denotes business supplies. Industry labels do not identify individual firm-level beneficiaries.

Part 4 Product innovation

Table A13. Daily risk metrics and coefficients of the NVDL-on-NVDA regression.
Cited in main text: pp. 5–6, section 4A–4C.

Asset	Daily volatility	95% VaR loss	95% CVaR loss
NVDA	3.746%	6.121%	7.073%
NVDL	7.479%	12.029%	14.162%

OLS term	Estimate	95% lower	95% upper
intercept	-0.000612	-0.000935	-0.000289
NVDA_slope	1.995975	1.987407	2.004543

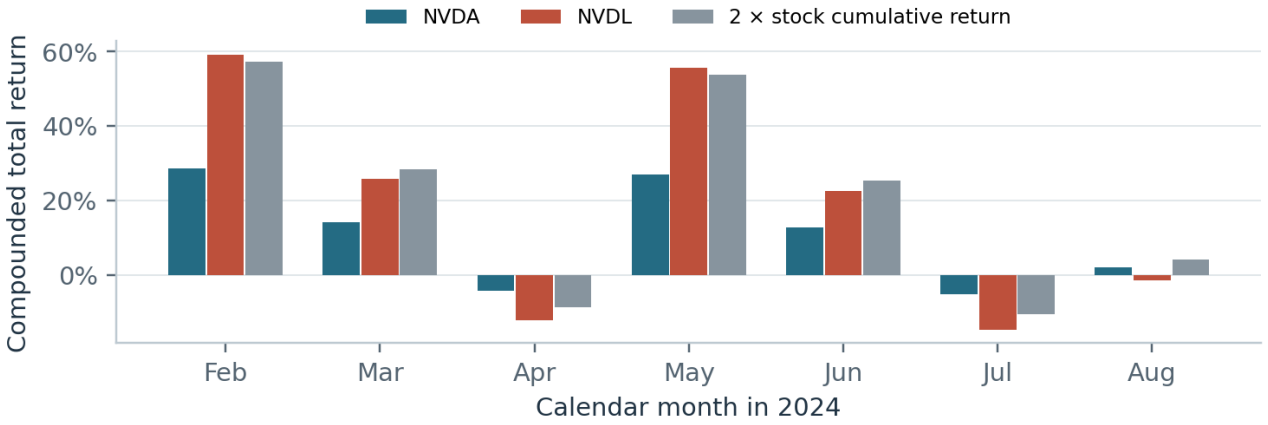
Source: instructor NVDA/NVDL daily total-return workbook, February 1–August 30, 2024; 147 paired US trading days. Sample standard deviations use n−1. VaR is minus the linearly interpolated 5th percentile; CVaR averages eight returns at or below that cutoff. Conventional OLS confidence intervals use Student t with 145 residual degrees of freedom.

Figure A10. NVDL daily returns against NVDA, with fitted and ideal daily 2× lines.
Cited in main text: p. 5, section 4B.



Same source and period as Table A13. Intercept × 252 = −15.41% per year, an arithmetic statistical approximation. The historical 1.15% net expense ratio comes from the assignment; the unexplained intercept cannot be attributed entirely to that fee.

Figure A11. Monthly cumulative returns and the comparison with twice NVDA's return.
Cited in main text: p. 6, section 4D and 4E.



Source: instructor NVDA/NVDL daily total returns, February–August 2024. Monthly returns compound daily observations; numerical results appear in Table A14.

Table A14. Monthly and full-period cumulative returns, with the two-day leverage example.
Cited in main text: p. 6, section 4D and 4E.

2024 period	Days	NVDA	NVDL	2 × NVDA
2024-02	20	28.60%	59.11%	57.19%
2024-03	20	14.21%	25.76%	28.42%
2024-04	22	-4.37%	-12.22%	-8.74%
2024-05	22	26.88%	55.72%	53.76%
2024-06	19	12.70%	22.60%	25.40%
2024-07	22	-5.28%	-14.68%	-10.56%
2024-08	22	2.01%	-1.60%	4.02%
Feb–Aug	147	94.05%	181.53%	188.10%

Hypothetical asset	Day one	Day two	Two-day return
NVDA	+10.00%	−9.00%	+0.10%
Ideal daily 2× ETF	+20.00%	−18.00%	−1.60%

Source: instructor NVDA/NVDL daily total returns, February–August 2024, and the assignment's hypothetical two-day example. Actual cumulative return = product of (1 + daily return) minus one. "2 × NVDA" doubles the stock's cumulative return for comparison; it is not the ETF's multi-day target. The ideal two-day ETF example ignores all fees and drag.

Part 5 Portfolio analytics

Table A15. MSCI Value factor regression coefficients and conventional OLS statistics.
Cited in main text: pp. 6–7, section 5B and 5C.

Term	Coefficient	OLS t	OLS p
Intercept	-0.000713	-0.906	0.366
RF	0.864636	2.788	0.006
Mkt_RF	0.948032	76.986	<0.001
HML	0.321642	19.682	<0.001
SMB	-0.174374	-9.905	<0.001

Source: classroom MSCI USA Standard Value Gross USD levels and French RF/Mkt−RF/HML/SMB. January 1995–July 2023, n = 343. Conventional OLS with an estimated RF coefficient and intercept; no Newey–West adjustment. R^2 = 0.9484. Intercept = −0.0713% monthly, −0.855% annualized; p = 0.366.

Figure A12. Cumulative wealth of MSCI Value and its factor replication.
Cited in main text: p. 7, section 5D.



Source: classroom MSCI USA Value Gross USD and French factors, January 1995–July 2023. Replication uses the full-sample estimates in Table A15.

Table A16. In-sample performance of MSCI Value and the replication excluding the intercept.
Cited in main text: p. 7, section 5D.

Portfolio	Annual mean	Annual volatility	Sharpe
Value	9.87%	15.27%	0.505
Replication	10.73%	14.87%	0.576

Same source and 343-month period. Replication excludes the regression intercept. Correlation with Value = 0.9739. Annual mean = 12 × monthly arithmetic mean; volatility = √12 × sample standard deviation; Sharpe subtracts annualized mean RF. Wealth starts at $1 on December 31, 1994. Coefficients are estimated and evaluated in the same sample; no implementation costs are deducted.

Table A17. Annualized active return, tracking error and information ratio versus MSCI USA.
Cited in main text: p. 7, section 5E.

Strategy	Annual active return	Annual TE	IR
Value	-1.343%	5.154%	-0.261
Minimum volatility	-0.796%	6.380%	-0.125
Equal weight	0.588%	5.396%	0.109
Momentum	2.158%	7.730%	0.279
Multifactor	0.152%	3.387%	0.045

Source: classroom MSCI USA Standard Gross USD indexes, January 1995–July 2023, n = 343. Each active return subtracts USA. Multifactor is the monthly average of four component returns. Annual active mean = 12 × monthly mean; TE = √12 × sample standard deviation; IR = annual active mean / TE. Returns exclude costs.

Figure A13. Component contributions to multifactor active-return variance.
Cited in main text: p. 8, section 5F.

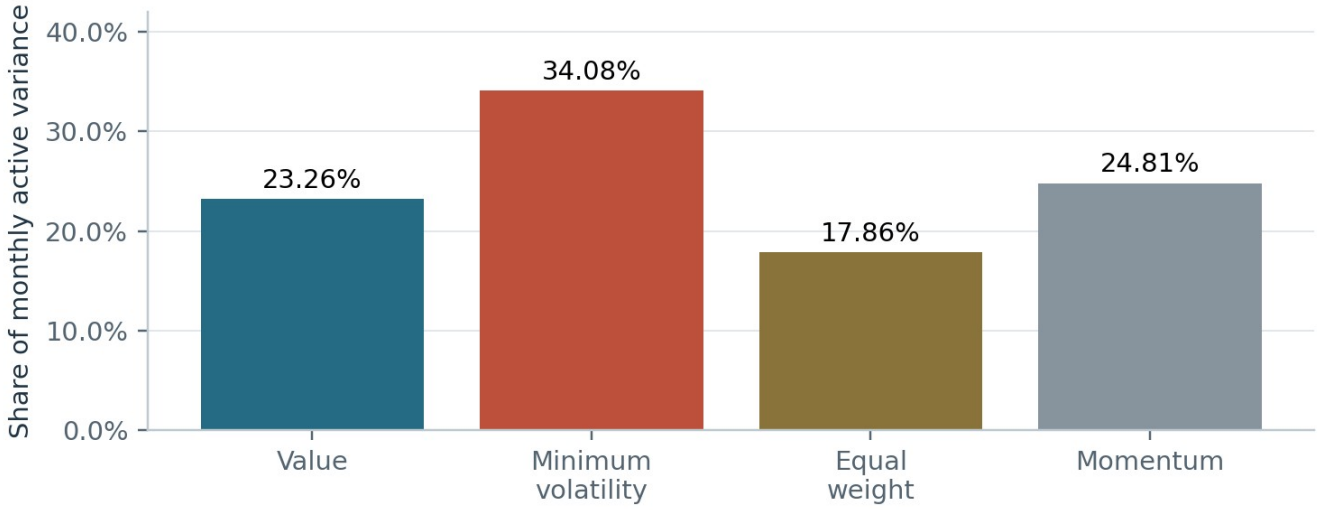


Table A18. Covariance contributions to active variance and annualized tracking error.
Cited in main text: p. 8, section 5F.

Component	Share of active variance	Annual TE contribution pp
Value	23.26%	0.788
Minimum volatility	34.08%	1.154
Equal weight	17.86%	0.605
Momentum	24.81%	0.840

Same source and period. With A = average(A_i), each monthly variance contribution is Cov(A, A_i)/4. Variance shares sum to 100% before rounding. TE contributions equal each share times total TE, summing to 3.387%; they are not standalone component tracking errors.

Sources and code

Table A19. Classroom input files and the historical windows used.
Cited in main text: p. 8, References.

Input	Use	Coverage used
ff_msci_shared.xlsx	Parts 1, 2 benchmark and 5	FF tests: 1927–Jul 2023; MSCI: Jan 1995–Jul 2023
security_characteristics.csv	Part 2	Cleaning: Nov 1998–Mar 2019 formations; backtest: 1999–2018 returns
industry_portfolios.xlsx	Part 3	Jan 1995–Jul 2023
product_innovation.xlsx	Part 4	Feb–Aug 2024 daily returns

Table A20. Original methodology sources and references supporting interpretation.
Cited in main text: p. 8, References.

Reference	Location
[1] French factors	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors.html
[1] Momentum deciles	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_10_port_form_pr_12_2.html
[1] Beta portfolios	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_port_form_BETA.html
[1] Industry portfolios	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_49_ind_port.html
[2] MSCI momentum	https://www.msci.com/eqb/methodology/meth_docs/MSCI_Momentum_Indices_Methodology.pdf
[3] MSCI value/growth	https://www.msci.com/eqb/methodology/meth_docs/MSCI_GIMIVGMethod_Feb2021.pdf
[4] Frazzini and Pedersen	https://www.aqr.com/Insights/Research/Journal-Article/Betting-Against-Beta
[5] OECD 2020	https://www.oecd.org/en/publications/oecd-digital-economy-outlook-2020_bb167041-en.html

Code and results

Code and results are available in the public project repository on GitHub. The source links in this report open the relevant files. Listing A1 contains the Part 3 optimization code. For setup and run commands, see the repository README.

Part 3A optimization code

Listing A1. Portfolio optimization module, lines 1–50 of 100. Continued on the following page.
Cited in main text: p. 4, section 3A.

Source: project src/part_3/portfolio_optimization.py. Original code; numerical calculations use full precision. Line numbering is for navigation and is not part of the source.

```
"""Part 3A: three industry allocation rules estimated as of December 2004."""
import numpy as np
import pandas as pd
from scipy.optimize import minimize

from .data_loading import (
    WEIGHT_TOLERANCE, monthly_window, output_directory, read_industry_returns,
    save_csv, save_json, software_versions, source_hash, validate_weights,
)


def sample_covariance(returns):
    covariance = returns.cov(ddof=1).to_numpy()
    if not np.isfinite(covariance).all() or not np.allclose(covariance, covariance.T):
        raise ValueError("Invalid sample covariance matrix.")
    if np.linalg.eigvalsh(covariance).min() <= 0:
        raise ValueError("Covariance is not positive definite; investigate the input.")
    return covariance


def minimum_variance(covariance, long_only=False):
    covariance = np.asarray(covariance, dtype=float)
    if covariance.ndim != 2 or covariance.shape[0] != covariance.shape[1]:
        raise ValueError("Covariance must be a square matrix.")
    if not np.isfinite(covariance).all() or not np.allclose(covariance, covariance.T):
        raise ValueError("Invalid covariance matrix.")
    if np.linalg.eigvalsh(covariance).min() <= 0:
        raise ValueError("Covariance must be positive definite.")
    n = len(covariance)
    ones = np.ones(n)
    if not long_only:
        solution = np.linalg.solve(covariance, ones)
        denominator = solution.sum()
        if not np.isfinite(denominator) or denominator <= 0:
            raise ValueError("Invalid minimum-variance normalization.")
        return solution / denominator, None
    annual = 12 * covariance
    result = minimize(
        fun=lambda weights: float(weights @ annual @ weights),
        x0=ones / n, jac=lambda weights: 2 * annual @ weights,
        method="SLSQP", bounds=[(0, None)] * n,
        constraints={"type": "eq", "fun": lambda weights: float(weights.sum() - 1),
                     "jac": lambda weights: ones},
        options={"ftol": 1e-12, "maxiter": 5000},
    )
    if not result.success:
        raise RuntimeError(f"Minimum-variance optimization failed: {result.message}")
    return result.x, result


```

Listing A1 continued

Source: project src/part_3/portfolio_optimization.py. Original code; numerical calculations use full precision. Line numbering is for navigation and is not part of the source.

```
 51  def highlight_extremes(column):
 52      values = column.drop(index="Total")
 53      styles = pd.Series("", index=column.index)
 54      bottom = values.sort_values(kind="stable").head(5).index
 55      top = values.sort_values(ascending=False, kind="stable").head(5).index
 56      styles.loc[bottom] = "background-color: #F8D7DA; color: #842029"
 57      styles.loc[top] = "background-color: #D1E7DD; color: #0F5132"
 58      return styles
 59
 60
 61  def run(root):
 62      training = monthly_window(read_industry_returns(root), "1995-01", "2004-12")
 63      covariance = sample_covariance(training)
 64      short, _ = minimum_variance(covariance)
 65      long_only, optimizer = minimum_variance(covariance, long_only=True)
 66      inverse = 1 / training.std(ddof=1).to_numpy()
 67      weights = pd.DataFrame({
 68          "min_var_short": short, "min_var_long_only": long_only,
 69          "inverse_volatility": inverse / inverse.sum(),
 70      }, index=training.columns).rename_axis("industry")
 71      validate_weights(weights, ("min_var_long_only", "inverse_volatility"))
 72      variances = [w @ covariance @ w for w in weights.to_numpy().T]
 73      if variances[0] > variances[1] + 1e-10 or variances[1] > variances[2] + 1e-10:
 74          raise ValueError("Unexpected in-sample variance ordering.")
 75      out = output_directory(root, "portfolio_optimization")
 76      save_csv(weights, out / "industry_weights_2004.csv")
 77      table = weights.copy()
 78      table.loc["Total"] = weights.sum()
 79      styled = table.style.set_uuid("industry_weights_2004").format("{:.2%}").apply(
 80          highlight_extremes, axis=0
 81      ).set_caption(
 82          "Industry weights as of December 31, 2004. Source: supplied Fama–French "
 83          "49-industry value-weighted monthly returns; estimation: January 1995–December 2004."
 84      )
 85      (out / "industry_weights_2004.html").write_text(styled.to_html(), encoding="utf-8")
 86      save_json({
 87          "source": "data/raw/industry_portfolios.xlsx",
 88          "source_sha256": source_hash(root, "data/raw/industry_portfolios.xlsx"),
 89          "worksheet": "49_Industry_Portfolios", "return_units": "Decimal",
 90          "return_type": "Value-weighted monthly total returns",
 91          "estimation_start": "1995-01", "estimation_end": "2004-12",
 92          "formation_date": "2004-12-31", "estimation_months": 120, "industries": 49,
 93          "covariance_ddof": 1, "solver": "SLSQP", "solver_ftol": 1e-12,
 94          "solver_maxiter": 5000, "solver_success": bool(optimizer.success),
 95          "solver_iterations": int(optimizer.nit), "weight_tolerance": WEIGHT_TOLERANCE,
 96          "highlight_ties": "Original industry order for exact ties",
 97          "software_versions": software_versions(),
 98      }, out / "optimization_metadata.json")
 99      print("Part 3A: 49 industries; three valid portfolios; estimation January 1995–December 2004.")
100      return weights
```